

Article

Enhancing Human–Robot Interaction: Development of Multimodal Robotic Assistant for User Emotion Recognition

Sergio Garcia , Francisco Gomez-Donoso *  and Miguel Cazorla 

University Institute for Computer Research, University of Alicante, P.O. Box 99 Alicante, Spain; sgm171@alu.ua.es (S.G.); miguel.cazorla@ua.es (M.C.)

* Correspondence: fgomez@ua.es

Abstract: This paper presents a study on enhancing human–robot interaction (HRI) through multimodal emotional recognition within social robotics. Using the humanoid robot Pepper as a testbed, we integrate visual, auditory, and textual analysis to improve emotion recognition accuracy and contextual understanding. The proposed framework combines pretrained neural networks with fine-tuning techniques tailored to specific users, demonstrating that high accuracy in emotion recognition can be achieved by adapting the models to the individual emotional expressions of each user. This approach addresses the inherent variability in emotional expression across individuals, making it feasible to deploy personalized emotion recognition systems. Our experiments validate the effectiveness of this methodology, achieving high precision in multimodal emotion recognition through fine-tuning, while maintaining adaptability in real-world scenarios. These enhancements significantly improve Pepper’s interactive and empathetic capabilities, allowing it to engage more naturally with users in assistive, educational, and healthcare settings. This study not only advances the field of HRI but also provides a reproducible framework for integrating multimodal emotion recognition into commercial humanoid robots, bridging the gap between research prototypes and practical applications.

Keywords: human–robot interaction; multimodal emotion recognition; social robotics; humanoid robots; natural language processing



Academic Editors: Kai Guo and Hongbo Yang

Received: 13 November 2024

Revised: 16 December 2024

Accepted: 18 December 2024

Published: 19 December 2024

Citation: Garcia, S.; Gomez-Donoso, F.; Cazorla, M. Enhancing Human–Robot Interaction: Development of Multimodal Robotic Assistant for User Emotion Recognition. *Appl. Sci.* **2025**, *14*, 11914. <https://doi.org/10.3390/app142411914>

Copyright: © 2024 by the authors. Licensee MDPI, Basel, Switzerland. This article is an open access article distributed under the terms and conditions of the Creative Commons Attribution (CC BY) license (<https://creativecommons.org/licenses/by/4.0/>).

1. Introduction

Humanoid robots represent a groundbreaking field with immense potential to redefine human–robot interaction (HRI), particularly in personal assistance. This potential is evident in applications such as healthcare, education, and assistive robotics [1]. A key aspect of making these robots effective, especially in domains such as healthcare, education, and assistive environments, is their capacity to perceive and respond to human emotions. The humanoid robot Pepper, developed by SoftBank Robotics (Tokyo, Japan), is a prime example of how these technologies are being used to create more empathetic and intuitive systems. Despite Pepper’s advanced capabilities for interaction, one of its main limitations lies in accurately interpreting and responding to human emotions.

This research sought to overcome these challenges by enhancing Pepper’s ability to recognize emotions through a multimodal approach. By combining data from visual, auditory, and textual inputs, the robot is able to gain a more nuanced understanding of human emotions, allowing for more accurate and contextually appropriate responses. Unlike previous studies, which often relied on single-modality approaches [2], our work integrates these data sources into a unified system, leveraging advanced techniques such

as convolutional neural networks (CNNs), transformers, and fine-tuning methods tailored to individual users.

The system's reproducibility is a core aspect of this study. Using Pepper, a commercially available robot, and open software tools, we provide a framework that can be replicated and adapted to different contexts. These advancements offer practical applications in domains such as healthcare, education, and social robotics, paving the way for robots to engage with users empathetically and effectively.

Such advancements in emotional recognition have significant implications for the future of assistive robotics. Through improving the human-likeness of robot interactions, robots can better fulfill roles that require emotional sensitivity, such as providing support to the elderly, educating children, or assisting individuals with disabilities.

This paper delves into the methods and technologies used to integrate multimodal emotional recognition into Pepper. It outlines the improvements in accuracy achieved through this approach and explores the impact of these enhancements on the robot's interactions. Furthermore, the challenges encountered during development and the solutions applied are discussed, providing valuable insights for future research in the field of HRI.

The remainder of this paper is organized as follows: Section 2 reviews related work in the field and highlights existing gaps. Section 4 describes the methodology and system components, focusing on the integration of multimodal emotion recognition. In Section 5, the experiments and results are presented, demonstrating the effectiveness of the proposed approach. Finally, Section 6 discusses the conclusions and potential directions for future research, particularly in the context of assistive and social robotics.

2. Related Works

This research intersects with several key areas of the state of the art. Below is a brief review of relevant studies to establish the necessary context.

On one hand, the field of assistive robotics has experienced significant growth, particularly with the development of humanoid robots like Pepper, which can perform both assistive tasks and serve as social companions. Research in this area has evolved from basic task automation to more complex interactions involving emotional and social intelligence, with the overarching goal of improving quality of life, especially for the elderly, children, or people with special needs [3]. A notable example is the advancement of Pepper as both an assistant for daily tasks and a social companion [1,4].

On the other hand, the area of human–robot interaction (HRI) has expanded significantly, driven by advances in AI and robotics, with a growing emphasis on the social capabilities of robots. The “Uncanny Valley” theory, introduced by Mori in the 1970s, continues to influence robot design, stressing the importance of balancing human-like features with user comfort [5]. Current studies on HRI aim to improve the robot's understanding of human behaviors and emotions, promoting more natural and intuitive interactions.

Neural networks play a central role in enabling robots to process and interpret large amounts of sensory data. Artificial neural networks (ANNs) and their subsets, such as convolutional neural networks (CNNs) and recurrent neural networks (RNNs), are essential for tasks ranging from visual recognition to the processing of sequential data like speech and text [6]. These technologies form the foundation of modern robotic systems, allowing significant advances in machine perception and decision making.

In recent years, transformer architectures have revolutionized natural language processing (NLP) by providing a mechanism for models to weigh the importance of different words in a sentence, regardless of their position. This approach has dramatically improved the performance of language models in tasks such as translation, sentiment analysis, and text generation [7]. In robotics, transformers are increasingly being used to enhance

robots' language comprehension and interaction capabilities, enabling more sophisticated dialogue systems that can engage in meaningful conversations with users [8].

Finally, the integration of multiple modalities—such as visual, auditory, and textual data—for emotion recognition represents a significant step forward in making robots more empathetic and responsive. This approach leverages the strengths of each modality to overcome the limitations of individual data types, enhancing the robot's ability to understand and respond to human emotional states comprehensively and accurately [9].

Despite these advances, current studies often focus on single-modality systems or specific use cases, limiting their adaptability in diverse environments [2]. This work sought to bridge these gaps by proposing a multimodal framework that integrates multiple sources of emotional data, enhancing both accuracy and robustness [10,11]. By addressing these limitations, our research provides a comprehensive and adaptable approach to advancing human–robot interaction.

3. System Hardware: The Pepper Humanoid Robot

Pepper is a semi-humanoid robot developed by SoftBank Robotics, designed primarily for human–robot interaction and social engagement [12]. Standing at approximately 1.2 m tall and equipped with a tablet on its chest, Pepper is intended to communicate with users through both verbal and nonverbal cues, making it suitable for a variety of domains such as healthcare, education, and customer service.

Unlike fully open source platforms, Pepper operates on a proprietary system called NAOqi OS, running on a Linux-based environment. This means that while the robot's hardware and lower-level firmware are not open source, developers can access a comprehensive set of Application Programming Interfaces (APIs) to extend Pepper's capabilities. SoftBank Robotics provides an official Software Development Kit (SDK) that primarily supports Python, though C++ and Java bindings are also available. The SDK includes modules for motion control, speech recognition, speech synthesis, face and person detection, and tactile sensor input, among other functionalities.

To integrate new capabilities into Pepper, developers typically connect to the robot's onboard system over a local network, using NAOqi APIs to control its sensors, actuators, and cognitive modules. This enables the implementation of custom behaviors, allowing researchers to modify Pepper's interactive capacities without altering its physical hardware. In our project, we leveraged these existing APIs to incorporate advanced emotion recognition modules and enhanced natural language processing. For the multimodal emotion recognition, the audio input from Pepper's microphones, the visual data from its cameras, and textual interpretation from spoken inputs were processed externally, and the results were fed back into Pepper via the NAOqi APIs. This approach lets other researchers reproduce our work by following the following steps:

1. Obtain a Pepper robot from SoftBank Robotics and ensure it is running the latest NAOqi OS version.
2. Install the official SDK and set up a development environment capable of connecting to Pepper's IP address via the local network.
3. Implement or integrate external modules (vision, speech recognition, NLP, emotion detection) in standard Python environments. These modules can run on a local machine or on a dedicated server.
4. Use the NAOqi APIs to send commands to Pepper. For example, developers can use the `ALMotion` module to control Pepper's movement, `ALTextToSpeech` for spoken output, `ALAudioDevice` for audio streaming, and `ALVideoDevice` for camera feed access.
5. Connect the emotion recognition and intention classification outputs from the external models back to Pepper's behaviors. For instance, if the system detects the user is sad,

Pepper can respond with a comforting remark using ALTextToSpeech, and adjust its posture to seem more empathetic via ALMotion.

By relying on Pepper's official SDK and NAOqi APIs, our approach is fully reproducible. Other researchers with access to Pepper can replicate and extend our system by following standard guidelines and utilizing the provided code, ensuring that the scientific knowledge generated is both accessible and verifiable.

4. Methodology

The proposed framework for enhancing Pepper's emotional recognition capabilities employs a comprehensive architecture that integrates visual, auditory, and textual data. This multimodal approach significantly improves Pepper's ability to understand and respond to human emotions, facilitating more accurate and empathetic interactions. This section is organized into several key components: an overview of the system architecture followed by detailed descriptions of each functional module.

4.1. System Architecture

The overall system architecture is meticulously designed to enable seamless interaction between the user and Pepper. The interaction process is initiated when the user records a voice message through a mobile application. When the user presses the recording button on the mobile interface, a signal is sent to activate Pepper's camera. The camera begins capturing frames of the user's facial expressions with a slight temporal delay, which are then passed through a CNN to analyze the predominant facial emotion over the recording duration. Simultaneously, the system records the audio message. Once the recording is complete, the audio is processed by a separate CNN to extract auditory emotions, while the text transcription of the audio is analyzed for textual emotion detection. These modalities are integrated to determine the overall emotional state of the user during the interaction. Once the recording is complete, the system processes the audio data, transcribed text, and visual inputs to ascertain the predominant emotion expressed by the user. The transcribed text undergoes intent analysis to determine the user's request, which then directs Pepper to perform corresponding actions such as setting reminders, controlling movement, initiating chatbot conversations, or activating object recognition functionalities. Figure 1 provides a visual representation of the system's architecture.

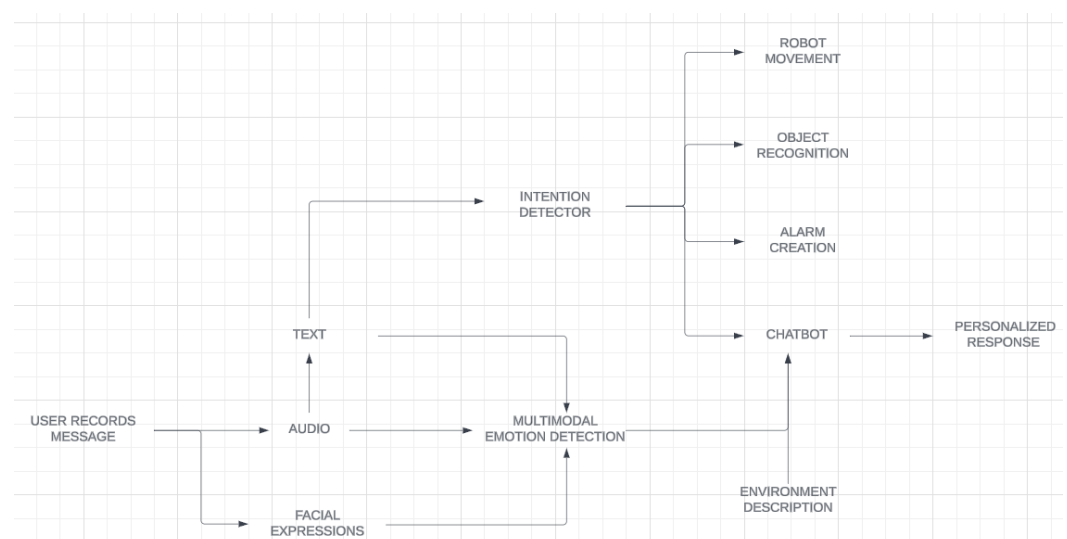


Figure 1. Interaction flow diagram.

4.2. Mobile Interface Development

The mobile interface comprises two primary components: a frontend mobile application developed using React Native and a backend server built on Node.js. The React Native application offers an intuitive and user-friendly interface for users to record and send voice messages. A single recording button facilitates the capture of audio, which is then transmitted to the backend server for processing.

The backend server leverages the Multer library for efficient file handling, receiving and temporarily storing the audio recordings sent from the mobile application. The audio files are converted and stored with specific parameters to ensure compatibility and quality for subsequent analysis. These parameters include the following:

- File Format: Conversion to WAV format to maintain compatibility with audio processing tools.
- Audio Codec: Utilization of the pcm_s16le codec to ensure high-fidelity, lossless audio encoding.
- Sampling Rate: Standardization of the sampling rate to 16,000 Hz.
- Audio Bitrate: Setting the bitrate to 256 kbps for superior audio quality.
- Audio Channels: Processing audio in mono to focus on voice analysis and reduce file size.

These specifications are carefully selected to match the original recording characteristics, ensuring accurate and consistent analysis.

4.3. Intention Recognition

The intention recognition module is responsible for interpreting user commands and determining the appropriate actions for Pepper to execute. This is accomplished using an RNN model developed with TensorFlow and Keras, trained on a labeled dataset consisting of phrases corresponding to various user intents. Each user phrase is first tokenized into numerical sequences, which are then transformed into feature vectors using an embedding layer. These feature vectors are processed by the RNN, which captures the sequential structure of language and classifies the user's intention.

This system allows Pepper to respond effectively to a wide range of natural language commands, enhancing the fluidity of human–robot interactions. Common intents recognized by the system include commands for movement, object detection, reminders, and conversation initiation.

Examples of typical user requests are as follows:

- Movement: "Move forward", "Rotate 90 degrees to the left."
- Object detection: "What do you see around you?", "Is there a table in the room?"
- Reminders: "Remind me to call John in 10 min.", "Set a reminder for my doctor's appointment."
- Conversation initiation: "Pepper, let's have a chat.", "Start a conversation mode."

As observed, key phrases such as "move", "turn", "remind", "object", or "conversation" help the model classify user requests efficiently. The network has been trained to recognize and categorize these linguistic patterns, enabling it to deduce the appropriate action without the need for rigid command syntax. This flexibility improves the natural flow of interaction between the user and Pepper, making the system more user-friendly and adaptable to different contexts.

The dataset for intention recognition was built manually due to the absence of an existing dataset suitable for our methodology. It includes balanced samples across the main categories of movement, reminders, object detection, and conversational commands. Each category contains 80 sentences, designed to cover diverse phrasing and scenarios.

The dataset was split into 75% for training and 25% for testing, ensuring robust evaluation. This approach ensures reproducibility and allows researchers to adapt or extend the dataset for similar studies.

4.4. Multimodal Emotion Recognition

This module integrates three distinct modalities—visual, auditory, and textual—to detect and classify user emotions into primary categories: angry, happy, neutral, and sad.

4.4.1. Visual Emotion Detection

Visual emotion detection utilizes a 2D-CNN to analyze grayscale video frames captured with Pepper's camera. The CNN architecture consists of multiple convolutional and pooling layers followed by dense layers to extract features and classify emotions. During interaction, the system continuously processes frames and averages the detected emotions over the duration of the audio message, providing a comprehensive emotional profile. The model is trained on combined datasets such as RAVDESS, CREMA-D, and the Christian Mejía dataset, augmented with personalized data to enhance accuracy [13–15].

4.4.2. Auditory Emotion Detection

Auditory emotion detection employs features like Mel-Frequency Cepstral Coefficients (MFCCs), Mel-Spectrograms, Chroma, and Spectral Contrast extracted using the Librosa library. A 1D-CNN processes these audio features to classify emotions from the speech signals. Data augmentation techniques, including speed variation and background noise addition, are applied to enhance model robustness. The model is trained on datasets such as RAVDESS, SAVEE, and AESDD, followed by fine-tuning on personalized data [14,16].

4.4.3. Textual Emotion Detection

Textual emotion detection involves transcribing audio to text, translating it into English using the MarianMT model from the Helsinki-NLP group [17], and applying a pretrained transformer-based model to classify the text into emotions like happiness, sadness, and anger. This ensures accurate sentiment analysis based on the content of the user's speech.

In our system, the outputs from the visual, auditory, and textual emotion recognition modules are processed simultaneously without the explicit weighting of one modality over another. Rather than computing a predominant emotion through a mathematical aggregation of these modalities, the system relies on the large language model (LLM) to interpret the results holistically. The LLM synthesizes the emotional data alongside contextual inputs from the scene descriptor to infer the most relevant emotional state in a given interaction.

This approach allows the system to dynamically prioritize modalities based on the situational context, leveraging the LLM's advanced reasoning capabilities. For example, in an emotionally ambiguous scenario, the scene descriptor and textual cues might influence the LLM to place greater emphasis on auditory inputs. This design avoids potential biases introduced by static weighting schemes and ensures more flexible and context-aware emotional recognition.

4.5. Scene Description

The Scene Description module utilizes Large Language and Vision Assistant (LLaVA), a sophisticated computer vision tool designed to combine visual input with language-based models for advanced scene analysis [18]. LLaVA leverages pretrained vision-language models to interpret complex visual environments, extracting high-level semantic details such as objects, scenes, and demographic attributes (e.g., age, gender, or activity). By

identifying key details such as location, gender, and age, LLaVA enables the chatbot to better understand the user's environment and adapt its responses accordingly. This ensures that the chatbot's interactions are more relevant and personalized based on the user's current situation.

The contextual information provided by LLaVA significantly impacts the chatbot's ability to generate meaningful responses. For instance, if a user asks for advice, the response would be different depending on the environment. In a park, the advice might be more casual and relaxed, whereas in a clinical setting, it could be more formal or medically oriented. The integration of LLaVA into our system allows it to handle diverse and dynamic scenarios, making the robot's interactions more adaptable and contextually aware.

To further illustrate, below are examples of how LLaVA interprets different visual environments and how this information is utilized by the chatbot to personalize its responses.

The following examples illustrate how LLaVA's analysis of scenes provides critical information for tailoring chatbot responses:

- Scene 1 (Figure 2a): The environment is a classroom, and the person detected is a female child. Based on this context, the chatbot might generate responses that are educational or academic in nature, or adapt to a communication style appropriate for a younger audience.
- Scene 2 (Figure 2b): The environment is a busy street, with buildings and vehicles in the background. The detected person is a young male, possibly an adolescent or young adult. In this context, the chatbot might provide advice relevant to an urban setting, such as navigating traffic or safety tips for a city environment.

By analyzing visual cues such as the person's surroundings and demographic information, LLaVA allows the chatbot to provide more relevant, context-aware responses. This capability enhances the overall interaction by making the chatbot's advice or guidance more tailored to the user's immediate environment.

It is important to note that the images in Figure 2 were generated using artificial intelligence. This decision was made to symbolically represent real-world scenarios while avoiding potential copyright issues associated with using actual photographs. These schematic illustrations are intended solely for explanatory purposes and do not impact the validity of the methodology, results, or conclusions presented in this study. All experiments and data processing were conducted with real-world interactions using the Pepper robot and actual user inputs.



Figure 2. Examples of different images to describe. (a) Student in a classroom. (b) Person in an urban street.

4.6. ChatBot Setup

The ChatBot component utilizes the Mixtral8x7b model, an open source language model [19]. It adapts to various interaction contexts by incorporating inputs like expected roles, environmental data from LLaVA, user emotions, and messages. The model processes this information to generate coherent and contextually appropriate responses. Communication between the local system and the server hosting the chatbot models and GPUs is managed via a RESTful API, ensuring efficient and seamless response generation [8].

4.7. Object Recognition

Object recognition functionality enables Pepper to identify and classify objects in real time using the YOLOv8 model [20]. The system captures images of the surrounding environment and processes them through YOLOv8 to detect and recognize objects based on predefined confidence thresholds. This module operates concurrently with other system functionalities, providing environmental awareness and enhancing user interaction by allowing Pepper to respond to object-related queries.

4.8. Alarm Creation

The Alarm Creation module allows users to set reminders through voice commands. The system interprets the time and message from the user's voice input, distinguishing between different time units such as seconds, minutes, and hours. It schedules reminders using a multithreaded approach to handle multiple alarms simultaneously, ensuring timely activation. When a reminder is due, Pepper announces the message, providing a timely notification to the user.

4.9. Social Robot Movement

The movement module of Pepper, the commercial humanoid robot used in our system, encompasses two core functionalities: person tracking and the execution of specific, user-defined movements. These capabilities enhance Pepper's ability to interact dynamically with its surroundings and respond to user commands in real time.

4.9.1. Person Tracking

Pepper employs advanced facial detection algorithms to follow a user by adjusting its position and orientation continuously. The tracking process relies on the detection of the three-dimensional coordinates $[x, y, z]$ of the user relative to Pepper's torso reference frame. This allows for the accurate calculation of both the distance and the orientation between Pepper and the tracked individual.

Pepper's movement is fine-tuned by constantly updating its position to maintain an optimal distance from the user, ensuring smooth and natural interaction. The tracking algorithm adapts the translation and rotation velocities based on the detected position, making real-time adjustments. The calculations for these adjustments are as follows:

- To adjust the distance, the translation speed is calculated as

$$vel_translation = \pm 0.1 + (position_X - threshold) \times 2 \quad (1)$$

- To adjust orientation, the rotation speed is calculated as

$$vel_rotation = \pm 0.1 + |position_Y - threshold| \quad (2)$$

In these formulas, the variable *threshold* represents the ideal distance Pepper tries to maintain from the user, and $[x, y, z]$ are the user's coordinates relative to Pepper's body. By keeping these values within an optimal range, Pepper ensures consistent and

reliable person tracking, allowing it to follow the user effectively without abrupt or unnatural movements.

4.9.2. Executing Specific Movements

In addition to tracking, Pepper is capable of performing precise movements based on verbal commands given by the user. This process involves several key steps that transform spoken instructions into executable actions for Pepper:

1. **Transcription:** The user's spoken command is first captured by Pepper's audio system and transcribed into text using built-in speech recognition technologies.
2. **Command Parsing:** The transcribed text is then analyzed and broken down into its components. The system identifies the action word (e.g., "move", "turn") and additional parameters such as direction (e.g., "left", "forward") and magnitude (e.g., distance in meters or angle in degrees).
3. **Action Execution:** After parsing the command, it is translated into executable instructions that are sent to Pepper's motion control system. These instructions are processed in real time, allowing Pepper to perform the desired movement with precision.

For example, if the user issues the command "Pepper, move forward two meters", the system will recognize the action ("move"), the direction ("forward"), and the distance ("two meters"). The parsed command is then converted into low-level control signals that guide Pepper to move forward by exactly two meters.

5. Experiments

In this section, we present the results obtained from the various tests and training sessions conducted during the development of our system. These experiments evaluate the performance of the models, including plots for accuracy and loss to track progress.

5.1. Intention Recognition

This subsection focuses on the training results of the RNN model used for intention recognition. The goal is to evaluate the model's effectiveness in correctly identifying user intentions, which is essential for seamless and efficient human–robot interaction. The accuracy/epochs curve is shown in Figure 3.

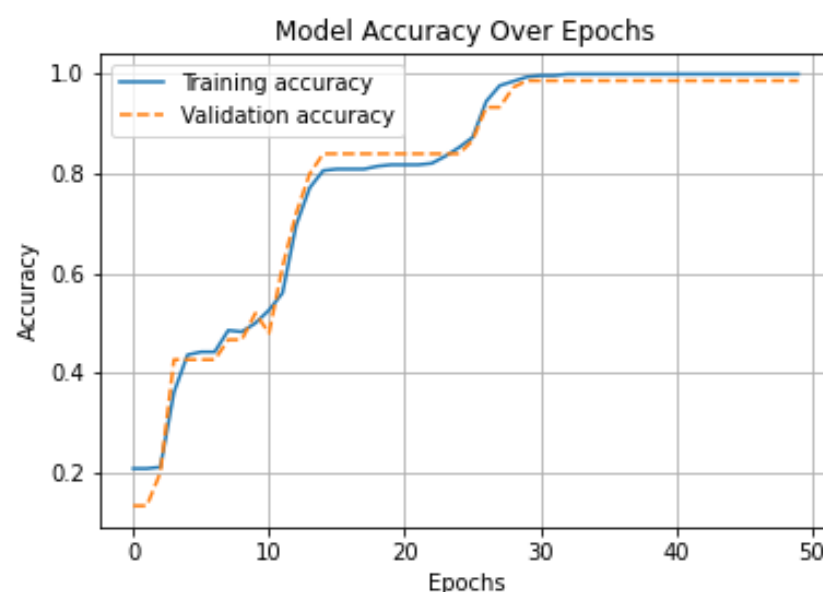


Figure 3. Accuracy graph of the intention classifier.

5.1.1. Dataset Description

The dataset used for training was built manually and includes the following categories: tracking, reminders, chatbot, movement, and object detection. Each category contains 80 sentences. The dataset was divided into 75% for training and 25% for testing to ensure a proper evaluation of the model.

5.1.2. Input and Output

The input to the RNN consists of tokenized user phrases, which are converted into numerical sequences. These sequences are processed through an embedding layer, transforming them into feature vectors. The output of the model is the classification of the phrase into one of the five intention categories.

A video showcasing the intention recognition system can be found at <https://youtu.be/ILoHKfFA4CY> (accessed on 16 December 2024).

5.1.3. Performance Analysis

The performance analysis of the accuracy and loss graphs demonstrates the model's effectiveness during training and validation phases. The accuracy curve shows a steady improvement, reaching values close to 100%, indicating the model's proficiency in learning to classify intentions correctly. The loss curve, on the other hand, steadily decreases, reflecting the model's improved capability to minimize errors.

5.1.4. Strengths and Limitations

The model performs well when the input phrases closely resemble those in the training dataset, as it effectively associates specific keywords with the appropriate intention class. However, there are limitations. The model struggles with handling complex, multi-module requests (e.g., "Tell me what objects are in the room and remind me to call later"). This limitation arises from the fact that the RNN is trained to classify single-intention commands, and it lacks the ability to process multiple intentions within a single input.

5.2. Emotion Recognition

This subsection outlines the experiments carried out to develop the multimodal emotion recognition module. As mentioned earlier, emotion detection in text relies on a pretrained model, so no experiments were performed specifically on textual emotion detection. A demonstration of the multimodal emotion recognition system can be viewed at <https://youtu.be/sWtVKhZLLXs> (accessed on 16 December 2024).

5.2.1. Datasets

To train the models for both audio and visual emotion recognition, we used carefully selected datasets known for their diversity and high quality in the field of artificial intelligence.

RAVDESS

The Ryerson Audio-Visual Database of Emotional Speech and Song (RAVDESS) contains controlled audiovisual recordings of actors performing a variety of emotional expressions [13]. For visual emotion recognition, we utilize the facial expression data, which include emotions like joy, sadness, anger, and surprise. On the audio side, the dataset provides high-quality recordings of emotionally expressive speech, which is ideal for training emotion recognition models based on voice analysis.

Crema-D

The CREMA-D (Crowd-sourced Emotional Multimodal Actors Dataset) offers both audio and video recordings of actors delivering emotionally charged dialogues [14]. The dataset stands out for its range of emotional expressions, as well as the diversity in the actors' backgrounds, making it particularly useful for training models that need to handle dynamic emotional expressions with varying intensity levels.

Christian Mejía Dataset

The dataset provided in [15] is a collection of facial images that represent a variety of emotions. This dataset pulls from well-known sources such as FER2013 and AFF-WILD and includes images generated by artificial intelligence, providing a rich resource for training facial emotion recognition models.

Surrey Audio-Visual Expressed Emotion (SAVEE)

Developed at the University of Surrey, the SAVEE dataset consists of high-quality voice recordings, specifically from native English-speaking males [16]. While this dataset is somewhat limited in terms of gender diversity, it offers recordings that span seven categories of emotions: anger, disgust, fear, happiness, sadness, surprise, and neutral.

Acted Emotional Speech Dynamic Database (AESDD)

The AESDD dataset is designed for speech emotion recognition (SER) and categorizes emotions into five types: anger, disgust, fear, happiness, and sadness. It is a valuable resource for training models focused on audio-based emotion detection [21].

5.2.2. Facial Emotion Detection

In this subsection, we analyze the performance of the models trained for facial emotion recognition across different neural network architectures and datasets. These experiments evaluate the accuracy and efficiency of each model in categorizing emotions correctly.

Several experiments were conducted with various architectures and datasets to determine the optimal model for facial emotion detection:

- **CLIP (Contrastive Language-Image Pre-training):** The CLIP model, developed by OpenAI, was tested for its capacity to recognize and classify emotions in facial images. The model was evaluated using several datasets, including RAVDESS, CREMA-D, and the Christian Mejía dataset. In these tests, the model performed well on classes such as "angry" and "happy", achieving accuracy rates between 70% and 90%, depending on the dataset. However, it struggled with "neutral" expressions, where the highest accuracy reached only 30%, likely due to the model's lack of training on neutral emotion labels. Accuracy for the "sad" class varied between 50% and 60%, depending on the dataset. We trained four predictive models: the base pretrained model and three fine-tuned models using different datasets. Three evaluation tests were performed—one for each dataset. The model fine-tuned with the Christian Mejía dataset achieved the highest accuracy, with an average of 75% across four emotions on RAVDESS and 47% on CREMA-D. Although the pretrained CLIP model could accurately associate images with clear emotional cues (e.g., tears indicating sadness), it struggled with more subtle emotional cues. Fine-tuning the model with additional datasets and introducing text labels like "A person expressing a neutral emotion" helped improve performance with neutral images. During evaluation, each image was classified by inputting four emotion labels, and the highest predicted value determined the final classification.

- **Visual Transformers (ViTs):** Visual Transformers (ViTs) were also tested, as these models can capture complex relationships between image features more efficiently. Due to hardware constraints, we initially tested the model on the RAVDESS dataset, where it achieved a validation accuracy of 95%, comparable to the 2D CNN model trained on the same dataset. Further testing on larger datasets using more computational resources is expected to improve robustness and generalization.
- **3D Convolutional Neural Networks (CNNs):** A 3D CNN architecture was used to process entire video sequences instead of individual frames, utilizing the RAVDESS dataset's video components. The goal was to determine if analyzing sequences could improve emotion detection compared to static frame-by-frame analysis. However, the 3D CNN model achieved only 70% accuracy on the validation set, which is significantly lower than the 95% accuracy obtained by the 2D CNN and Visual Transformers. Due to this underperformance, the 3D CNN architecture was not pursued further.
- **2D Convolutional Neural Networks (CNNs):** Separate 2D CNNs were initially trained for each dataset (Crema-D, RAVDESS, and Christian Mejía). Datasets without pre-defined training and validation splits were manually segmented, with 75% of the data used for training and 25% for validation. Subsequently, data augmentation techniques were applied to the training sets to assess their impact on the model's generalization capabilities.

The main objective of these experiments was to assess each model's performance across different datasets and identify which emotions were more easily detected and which presented greater challenges.

Following extensive testing, the 2D CNN model emerged as the most effective. The final experiment involved combining all datasets and fine-tuning the model with personalized data. The performance graphs for this final model are shown in Figure 4.

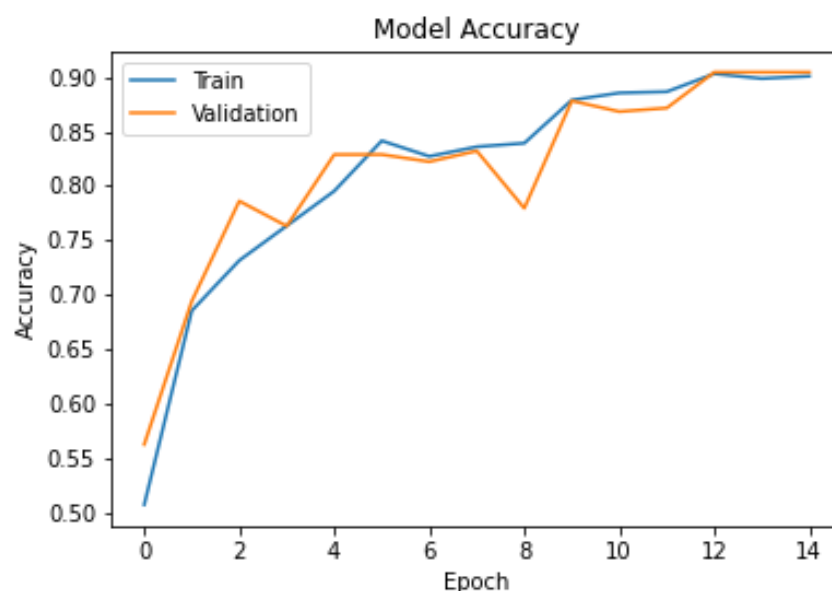


Figure 4. Accuracy graph of the fine-tuned facial emotion recognition model using the personalized dataset.

Although the fine-tuning was performed for a single person, achieving an accuracy close to 100% is not always possible, and there is a simple explanation for this. From each user-recorded message, five frames per second were extracted. This means that even if the user was angry throughout the message, not all frames would necessarily show micro-expressions of anger; some might appear neutral. In practice, this does not present a

significant problem because the system averages the detected emotions over the entire sequence. For example, if a seven-second conversation detects 76% anger and 24% neutrality, the system concludes that the user was predominantly angry while conveying the message.

5.2.3. Auditory Emotion Detection

In this subsection, we evaluate the performance of the auditory emotion detection model, which employs a 1D convolutional neural network (1D CNN). The main objective is to assess the model's ability to accurately predict emotions from audio data, using metrics such as accuracy and loss plots. We also analyze which emotions are easier or more difficult for the model to classify. Additionally, we compare the model's performance across various datasets, both individually and in combination, with and without the application of data augmentation techniques. This evaluation will help identify the best data configuration for training and guide the fine-tuning process.

In the final experiment, we combined all datasets with data augmentation and conducted fine-tuning using a personalized dataset.

The results of these experiments show a significant improvement after fine-tuning, with the model achieving an accuracy of 100%. However, it is important to highlight that the effectiveness of the fine-tuning process is closely tied to the user's ability to distinguish and express different emotions clearly in the training examples. The aim is to enable users to create their own personalized dataset in approximately 5 min. However, due to the limited amount of data in this short recording period, the system may be sensitive to factors like microphone distance or changes in the user's vocal tone from day to day.

Ideally, the system would benefit from a more extended data collection period. For instance, if a user could record four messages each day over the course of a week—one for each emotion, while varying their tone and the distance from the microphone slightly—the model would become more robust to such variations. Nonetheless, the system performs well under stable conditions, provided that there are no dramatic shifts in tone or microphone positioning.

A video demonstrating an interaction between the user and the robot can be found at the following link: <https://www.youtube.com/watch?v=dKuLXqDm8v8> (accessed on 16 December 2024).

5.3. Discussion and Comparative Analysis

The results of our experiments demonstrate the effectiveness of our proposed multimodal emotion recognition system, particularly when applied to human–robot interaction (HRI). The integration of visual, auditory, and textual modalities has led to improved accuracy and robustness in recognizing user emotions, as shown in Figures 4 and 5. For instance, our visual emotion detection model achieved a fine-tuned accuracy of approximately 95%, while the auditory model reached 100% accuracy under controlled conditions. However, these results must be interpreted in the context of their respective limitations.

Compared to prior studies, such as [10], which utilize fixed datasets for emotion recognition, our approach demonstrates the adaptability of a multimodal system through the application of fine-tuning. This fine-tuning is key to tailoring the system to specific users, addressing the inherent variability in emotional expression between individuals. Unlike previous work, where general models are applied across datasets without user-specific adjustments, our methodology enables explicit personalization, making it particularly suitable for domains like healthcare and education. However, this advantage comes with challenges, including the need for user-specific data collection and a dependency on stable environmental conditions during the fine-tuning process.

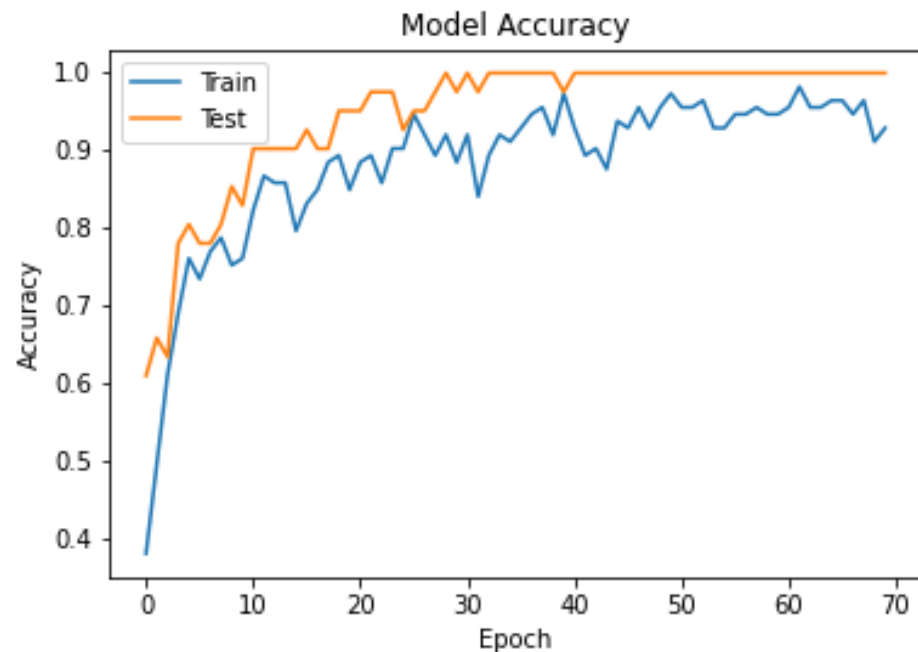


Figure 5. Accuracy graph of the fine-tuned auditory emotion recognition model with the personalized dataset.

Moreover, while our system achieves high accuracy in emotion recognition, it currently lacks a fully automated mechanism for balancing the weight of individual modalities (e.g., visual versus auditory emotions) in ambiguous cases. This limitation is an area of active research and presents opportunities for future advancements in multimodal systems.

Another notable strength of our approach is its ability to integrate seamlessly with Pepper's existing framework, demonstrating real-world applicability. Unlike prior works that focus primarily on simulation or controlled settings, our results indicate the feasibility of deploying such systems in practical scenarios. Nevertheless, the reliance on Pepper's outdated hardware and limited computational power poses challenges for real-time operation and scalability, which we aim to address in future iterations of the system.

In conclusion, the findings of this study not only validate the proposed framework but also highlight areas for further development, including hardware improvements, the automation of fine-tuning processes, and broader generalization to diverse user groups. This discussion underscores the potential impact of multimodal emotion recognition in advancing the field of HRI while acknowledging the scientific and practical challenges that remain.

6. Conclusions and Future Work

This study addresses several fundamental challenges in human–robot interaction (HRI), particularly in the domains of emotional understanding, contextual reasoning, and decision making. By integrating multimodal emotion recognition capabilities into the Pepper humanoid robot, our approach combines visual, auditory, and textual data to provide a more holistic understanding of user emotions. This enables Pepper to respond with greater sensitivity and contextual awareness, representing a significant advancement in creating robots capable of empathetic and meaningful interactions.

Compared to previous studies, which often rely on single-modal approaches or focus on individual components such as speech or facial recognition, our work introduces a unified framework that synthesizes data from multiple sources. This integration not only enhances the robustness and adaptability of emotion recognition systems but also

addresses broader scientific problems related to the perception and reasoning capabilities of robots. By leveraging an LLM, the system dynamically interprets multimodal inputs and contextual cues, demonstrating the potential for robots to act empathetically and intelligently in real-world scenarios.

Despite these advancements, several limitations must be acknowledged. First, the hardware constraints of Pepper, including outdated sensors and limited computational power, pose challenges to achieving real-time performance and scalability. Second, the process of integrating new users requires manually creating personalized datasets, which can be labor-intensive and difficult for users to replicate, especially when mimicking specific emotions. Lastly, while the system performs well with mid-range devices, its generalizability to broader demographics and diverse use cases remains to be tested. These limitations highlight areas for improvement and guide future research directions.

To address these shortcomings, future work should focus on integrating the pipeline into more advanced humanoid robots equipped with state-of-the-art sensors and enhanced hardware capabilities. Additionally, incorporating automated dataset generation tools and leveraging hybrid datasets—combining personalized and standardized data—could improve both scalability and generalization. The adoption of advanced vision and language models, such as large language models (LLMs) and more sophisticated scene description systems, will also be crucial for maintaining state-of-the-art performance as technology evolves.

Furthermore, this study contributes to understanding how robots can use emotional information to guide decision making, such as by adjusting tone, posture, or conversational content based on a user's emotional state. By addressing the complexity of recognizing and reasoning about human emotions in context, this work highlights the broader scientific implications for designing robots capable of human-like interaction.

In conclusion, this study lays the groundwork for the next generation of emotionally intelligent robots, bridging the gap between human-like interaction and real-world applicability. The integration of multimodal emotion recognition not only advances the current state of HRI but also underscores the potential of robots to become empathetic collaborators in diverse environments. As the field of humanoid robotics continues to evolve, the innovations demonstrated in this work represent a critical step toward a future where robots seamlessly coexist with humans, offering meaningful and personalized support in daily life.

Author Contributions: Conceptualization, M.C. and F.G.-D.; methodology, F.G.-D.; software, S.G.; validation, S.G.; formal analysis, F.G.-D.; investigation, F.G.-D. and S.G.; resources, M.C.; writing—original draft preparation, S.G. and F.G.-D.; writing—review and editing, F.G.-D. and M.C.; supervision, M.C.; project administration, F.G.-D.; funding acquisition, M.C. All authors have read and agreed to the published version of the manuscript.

Funding: This research was funded by the Spanish Ministry of Science, Innovation and Universities under grant PID2022-138453OB-I00.

Data Availability Statement: The data presented in this study are openly available in Zenodo at <https://doi.org/10.5281/zenodo.1188976>, in GitHub at <https://github.com/CheyneyComputerScience/CREMA-D>. Other datasets do not have DOI, but are publicly available in <http://kahlan.eps.surrey.ac.uk/savee/Download.html> (accessed on 16 December 2024) and <https://m3c.web.auth.gr/research/aesdd-speech-emotion-recognition/> (accessed on 16 December 2024).

Conflicts of Interest: The authors declare no conflicts of interest.

References

1. Wu, X.; Bartram, L. Social robots for people with developmental disabilities: A user study on design features of a graphical user interface. *arXiv* **2018**, arXiv:1808.00121.
2. Scherer, K.; Bänziger, T.; Roesch, E. *Blueprint for Affective Computing: A Sourcebook*; Oxford University Press: Oxford, UK, 2010.
3. Marin, E.G.C.; Morales, C.A.; Sanchez, E.S.; Cazorla, M.; Plaza, J.M.C. Designing a cyber-physical robotic platform to assist speech-language pathologists. *Assist. Technol.* **2023**, *35*, 48–55. [\[CrossRef\]](#) [\[PubMed\]](#)
4. Costa, A.; Martinez-Martin, E.; Cazorla, M.; Julian, V. PHAROs—PHysical Assistant RObot system. *Sensors* **2018**, *18*, 95–107. [\[CrossRef\]](#) [\[PubMed\]](#)
5. Kanda, T.; Miyashita, T.; Osada, T.; Haikawa, Y.; Ishiguro, H. Analysis of humanoid appearances in human–robot interaction. *IEEE Trans. Robot.* **2008**, *24*, 725–735. [\[CrossRef\]](#)
6. Rumelhart, D.E.; Hinton, G.E.; Williams, R.J. Learning representations by back-propagating errors. *Nature* **1986**, *323*, 533–536. [\[CrossRef\]](#)
7. Vaswani, A.; Shazeer, N.; Parmar, N.; Uszkoreit, J.; Jones, L.; Gomez, A.N.; Kaiser, L.; Polosukhin, I. Attention is all you need. *arXiv* **2023**, arXiv:1706.03762.
8. Raffel, C.; Shazeer, N.; Roberts, A.; Lee, K.; Narang, S.; Matena, M.; Zhou, Y.; Li, W.; Liu, P.J. Exploring the limits of transfer learning with a unified text-to-text transformer. *J. Mach. Learn. Res.* **2020**, *21*, 1–67.
9. Liang, C.; Lu, J.; Yan, W.Q. Human action recognition from digital videos based on deep learning. In Proceedings of the 5th International Conference on Control and Computer Vision, Xiamen, China, 19–21 August 2022; pp. 150–155.
10. Adiga, S.; Vaishnavi, D.; Saxena, S.; Tripathi, S. Multimodal Emotion Recognition for Human Robot Interaction. In Proceedings of the 2020 7th International Conference on Soft Computing & Machine Intelligence (ISCMI), Stockholm, Sweden, 14–15 November 2020; pp. 197–203. [\[CrossRef\]](#)
11. Liu, W.; Zheng, W.L.; Lu, B.L. Emotion Recognition Using Multimodal Deep Learning. In *Neural Information Processing, Proceedings of the 23rd International Conference on Neural Information Processing (ICONIP), Kyoto, Japan, 16–21 October 2016*; Lecture Notes in Computer Science; Springer: Cham, Switzerland, 2016; Volume 9948, pp. 511–519. [\[CrossRef\]](#)
12. Robotics, S. Pepper Robot Documentation. 2023. Available online: http://doc.aldebaran.com/2-5/home_pepper.html (accessed on 11 December 2024).
13. Livingstone, S.R.; Russo, F.A. The Ryerson Audio-Visual Database of Emotional Speech and Song (RAVDESS): A dynamic, multimodal set of facial and vocal expressions in North American English. *PLoS ONE* **2018**, *13*, e0196391. [\[CrossRef\]](#) [\[PubMed\]](#)
14. Cao, H.; Cooper, D.G.; Keutmann, M.K.; Gur, R.C.; Nenkova, A.; Verma, R. Crema-d: Crowd-sourced emotional multimodal actors dataset. *IEEE Trans. Affect. Comput.* **2014**, *5*, 377–390. [\[CrossRef\]](#) [\[PubMed\]](#)
15. Mejia-Escobar, C.; Cazorla, M.; Martinez-Martin, E. A Large Visual, Qualitative, and Quantitative Dataset for Web Intelligence Applications. *Comput. Intell. Neurosci.* **2023**, *2023*, 1094823. [\[CrossRef\]](#)
16. Jackson, P.; Haq, S. *Surrey Audio-Visual Expressed Emotion (SAVEE) Database*; University of Surrey: Guildford, UK, 2014.
17. Tiedemann, J.; Thottingal, S. OPUS-MT—building open translation services for the world. In Proceedings of the 22nd Annual Conference of the European Association for Machine Translation, Lisboa, Portugal, 3–5 November 2020; pp. 479–480.
18. Liu, H.; Li, C.; Wu, Q.; Lee, Y.J. Visual instruction tuning. *Adv. Neural Inf. Process. Syst.* **2024**, *36*, 34892–34916.
19. Jiang, A.Q.; Sablayrolles, A.; Mensch, A.; Bamford, C.; Chaplot, D.S.; Casas, D.d.l.; Bressand, F.; Lengyel, G.; Lample, G.; Saulnier, L.; et al. Mistral 7B. *arXiv* **2023**, arXiv:2310.06825.
20. Khalaf, A.L.; Abdulrahman, M.M.; Al Barazanchi, I.I.; Tawfeq, J.F.; JosephNg, P.S.; Radhi, A.D. Real time pedestrian and objects detection using enhanced YOLO integrated with learning complexity-aware cascades. *TELKOMNIKA Telecommun. Comput. Electron. Control.* **2024**, *22*, 362–371. [\[CrossRef\]](#)
21. Vryzas, N.; Kotsakis, R.; Liatsou, A.; Dimoulas, C.A.; Kalliris, G. Speech emotion recognition for performance interaction. *J. Audio Eng. Soc.* **2018**, *66*, 457–467. [\[CrossRef\]](#)

Disclaimer/Publisher’s Note: The statements, opinions and data contained in all publications are solely those of the individual author(s) and contributor(s) and not of MDPI and/or the editor(s). MDPI and/or the editor(s) disclaim responsibility for any injury to people or property resulting from any ideas, methods, instructions or products referred to in the content.